Low Dropout Regulator

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1 Introduction

This project is about a Low Dropout Voltage Regulator. This circuit supplies a stable voltage of 1.6V with a power supply of 1.8V. This supplies a current of 1mA for a standard load of 1.6kOhms.

2 Design Process

2.1 Design of OpAmp

For our PSRR requirements, we need an op-amp with a gain of over 60 dB. This means a single-stage OTA will not suffice, we were left with having to decide between a two-stage OTA, single-stage cascode opamp or single stage folded cascode opamp. The cascode was infeasible because of the 1.8V VDD having to be distributed across 6 levels, the folded cascode offered design challenges, hence we decided on the two-stage ota with appropriate frequency compensation.

First, we started with the Differential Pair Transistors. We decided on a current of 100μA to flow through the two transistors together. This makes 50μA flow through each individual transistor.

The voltage reference we use for our LDO comes from a bandgap voltage reference of 1.23 volts, hence the common mode of our op-amp should be configured as such.

Start off with giving all three stages 200 mV of overdrive. We see by dc analysis in this case that there exists an ICMR of +/- 300mV from the common mode of 1.23 volts. This is sufficient for the purposes of our requirement for the op amp, but the ICMR can be further optimized by varying the overdrives given to each of the levels. The upper pmos current mirror level will always remain in saturation because of the way it is connected, with the drain voltage always being equal to the gate voltage. Hence, the Vds provided to it can be minimized.

Once we got to know the overdrives and currents, we found the $\frac{W}{L}$'s for each of the transistors in the first stage ota. The W/L's were then tweaked to reduce the sizes of the transistors for easier layout and for better ICMR.

Once the single stage OTA was functioning as expected, We added the second stage. The overdrives were again assumed to be 150 mV and the current was taken as 3 times the tail current of the first stage after which the $\frac{W}{L}$'s was calculated.

We expected to drive a load of 5 pF, which is the capacitance of the pass transistor PMOS as seen from its gate; hence, we set the compensation capacitor to 1.2 pF and the zero resistor of value 450 . This was followed by stability and dc analysis.

We found the op-amp to have gain of about 60 dB and a phase margin of about 60° with a gain bandwidth product of 100 MHz. These specifications will be sufficient for the LDO that we seek to design.

2.2 Design of LDO

Our specifications dictate that our LDO flows a current of 1mA for a voltage drop of 200mV. Since this has to be greater than the pass transistor's overdrive, we size it assuming it to have an overdrive of 100 mV. This leaves us with deciding the resistor values and the frequency compensation means.

We allow 20 μA current to flow through the resistors, we know the voltage drop and the currents across the resistors from which we can calculate the magnitudes of the resistances. This turns out to be 18.5 k and 61.5 k.

Now when we checked the stability criteria, there were two poles and the phase margin was -20degrees. So we tried identifying the pole location and cancelled the pole by adding a zero with the help of ESR of capacitor. And ESR of 100 was chosen which yielded a good result of 45 degrees of Phase Margin.

Now we tried checking the load regulation of the LDO. And we found that a resistor of $3k$ gives the best phase margin and the worst phase margin was obtained when a resistor of 500 was connected to the output of the LDO. The variation in phase margin was from 60° to 40°.

2.3 Post Layout Corner Simulations

There are two kinds of simulation, stability analysis and Transient Analysis. The Stability analysis gives information about the loop gain and the Phase Margin of the system.

2.3.1 FS corner

Stability and Transient Analysis

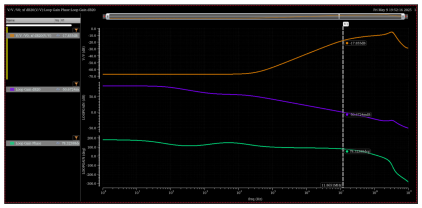


Figure 1: Stability Analysis

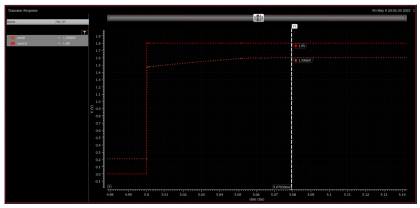


Figure 2: Transient Analysis

2.3.2 FF corner

Stability and Transient Analysis

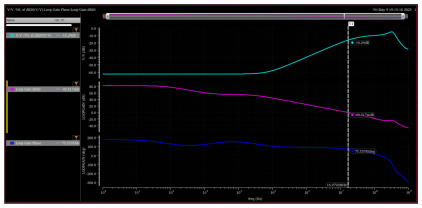


Figure 3: Stability Analysis

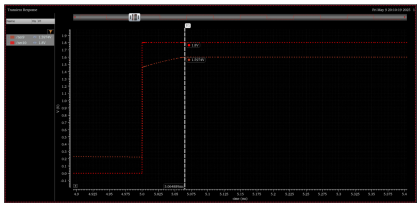


Figure 4: Transient Analysis

2.3.3 SS corner

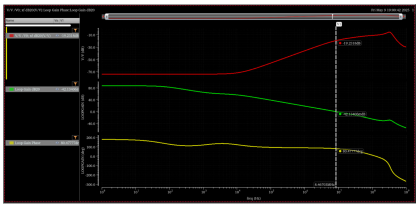


Figure 5: Stability Analysis

2.3.4 TT corner

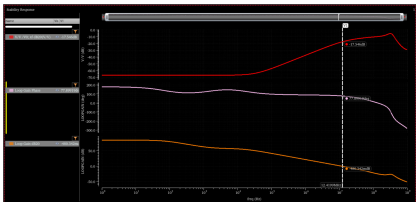


Figure 6: Stability Analysis

2.3.5 SF corner

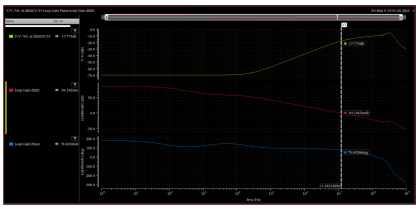


Figure 7: Stability Analysis

2.4 Layout view

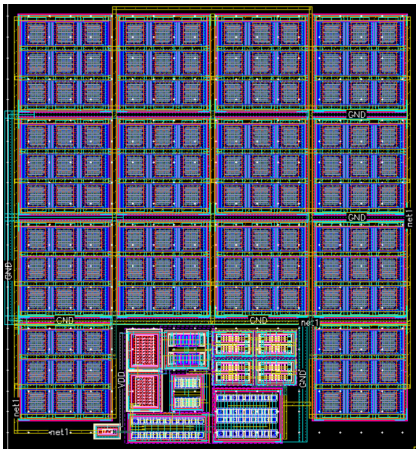


Figure 8: OTA

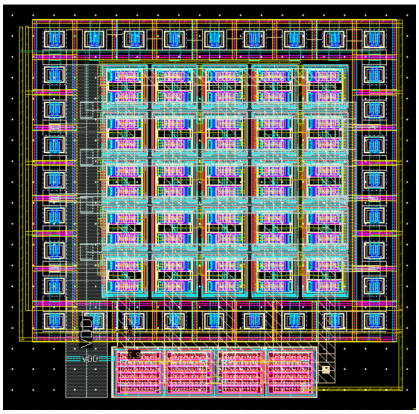


Figure 9: Pass Transistor

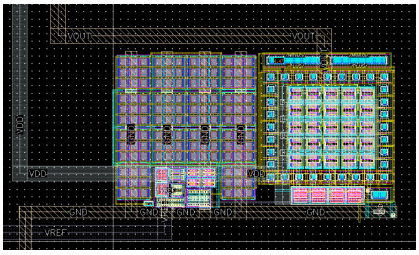
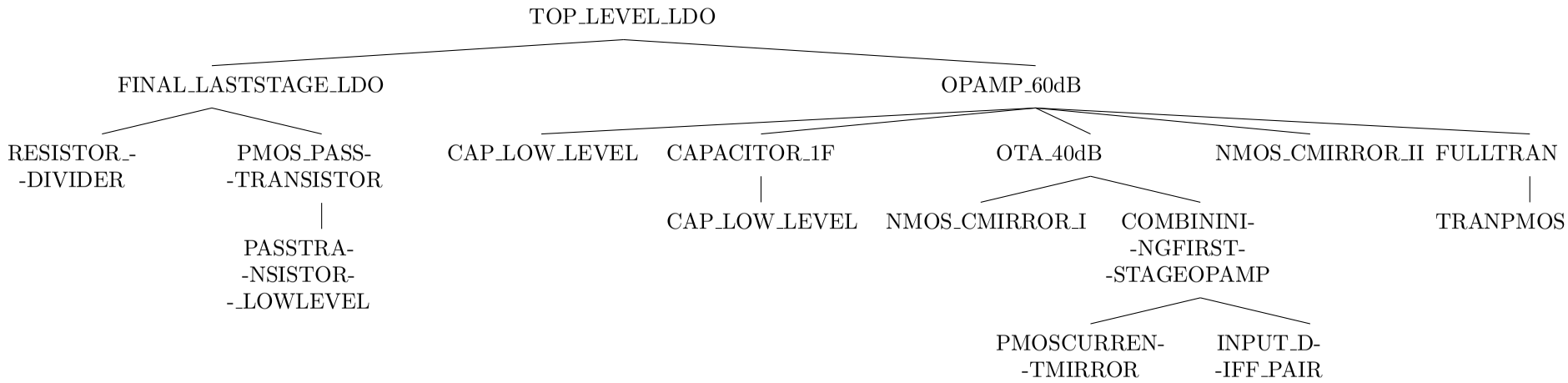


Figure 10: LDO

2.5 Hierarchy of files


TOP_LEVEL_LDO
FINAL_LASTSTAGE_LDO
OPAMP_60dB
RESISTOR_-
-DIVIDER
PMOS_PASS-
-TRANSISTOR
PASSTRA-
-NSISTOR-
-_LOWLEVEL
CAP_LOW_LEVEL
CAPACITOR_1F
CAP_LOW_LEVEL
OTA_40dB
NMOS_CMIRROR_I
COMBININI-
-NGFIRST-
-STAGEOPAMP
PMOSCURREN-
-TMIRROR
INPUT_D-
-IFF_PAIR
NMOS_CMIRROR_II
FULLTRAN
TRANPMOS

2.6 Test Circuits

There are three test circuits in the file:

- TEST_CIRCUIT
- LDO_TEST
- LDO_TEST_II